

Sarat Sindhu Mukhopadhyay

PHD STUDENT · QUALITY, RELIABILITY AND OPERATIONS RESEARCH

Indian Statistical Institute, Bangalore

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Education

Indian Statistical Institute, Bangalore

PHD IN QUALITY, RELIABILITY AND OPERATIONS RESEARCH

- (Thesis Submitted)
- Advisor: Dr. Gijo E. V.

Bangalore, Karnataka, India

Probable completion: 2026

Indian Statistical Institute, Bangalore

MS IN QUALITY MANAGEMENT SCIENCE

- Master's Thesis Advisor: Dr. B. S. Daya Sagar

Bangalore, Karnataka, India

2019

Presidency University, Kolkata

BSC. IN STATISTICS (HONS.)

Kolkata, West Bengal, India

2017

Area of Interests

My research interests broadly focus on statistical modeling and inference for life-testing and reliability models, particularly *Step-Stress Accelerated Life Testing (SSALT)* under non-constant hazard rate settings. I have worked on reliability modeling under different experimental and censoring schemes, including progressive censoring, repeated Type-II censoring, and competing risks, with an emphasis on dependent failure mechanisms. Currently, I am working on *degradation modeling under stressed conditions* for reliability assessment. I am also exploring *copula-based models* for dependence modeling in reliability and other statistical applications, as well as *frailty models* for complex reliability testing and heterogeneous populations.

Publications

PUBLISHED

Sarat Sindhu Mukhopadhyay, Ayan Pal, and E. V. Gijo. "Analysis of multiple step-stress dependent competing risks under proportional hazard model." *Quality Technology & Quantitative Management*, 23.3 (2026), pp. 475–504.

Sarat Sindhu Mukhopadhyay, K. V. Viswakala, and E. V. Gijo. "Inferences on Lifetime Performance Index of Weibull Products Under Step-Stress Setup." *IEEE Transactions on Reliability*, 74.4 (2025), pp. 4870–4884.

Sarat Sindhu Mukhopadhyay, Ayan Pal, and E. V. Gijo. "A failure-rate based multiple step-stress dependent competing risks model with applications to aerospace electrical connectors data." *Journal of Applied Statistics*, 2026, pp. 1–30.

IN REVIEW

Sarat Sindhu Mukhopadhyay, and E. V. Gijo "Inference for Proportional Hazard Step-Stress Data Using Inverse Estimation and Fiducial Methods."

Sarat Sindhu Mukhopadhyay, Ayan Pal, Debasis Kundu, and E. V. Gijo. "Time-Shifted Failure Rate-Based Modeling for Step-Stress Accelerated Life Testing."

Professional Experience

Feb 2019 – Jun 2019 **Data Scientist Intern**, Cummins Inc., Pune, India

May 2018 – Jul 2018 **Research and Development Intern**, Indian Market Research Bureau (KANTAR IMRB International), Kolkata, India

Awards & Recognitions

- 2024 **Prof. U.S. Nair Best Paper Presentation Award**, Tenth International Conference on Statistics for the Twenty-First Century (ICSTC 2024)
- 2025 **Professor Miss Aleyamma George Best Paper Award**, Eleventh International Conference on Statistics for the Twenty-First Century (ICSTC 2025)

Contributed Presentations (Conference and Workshop)

Sarat Sindhu Mukhopadhyay, and Gijo E. V. “ α Shaped Exponential Entropy.” Oral presentation, *International Workshop on Reliability Theory and Survival Analysis (IWRTSA 2022)*, organized by Indian Institute of Science Education and Research (IISER) Kolkata, India.

Sarat Sindhu Mukhopadhyay, Ayan Pal, and Gijo E. V. “Inference Based on a Novel Failure-Rate Based Multiple Step-Stress Dependent Competing Risks Model.” Oral presentation, *Tenth International Conference on Statistics for the Twenty-First Century (ICSTC 2024)*, organized by the International Statistics Fraternity (ISF) in collaboration with the Department of Statistics and the School of Physical and Mathematical Sciences, University of Kerala, and the American Statistical Association (ASA), India.

Sarat Sindhu Mukhopadhyay, Viswakala K. V, and Gijo E. V. “Inferences on Lifetime Performance Index of Weibull Products under Step-Stress Setup.” Oral presentation, Annual Conference of the International Indian Statistical Association (IISA 2024), Cochin University of Science and Technology, India.

Sarat Sindhu Mukhopadhyay, and Gijo E. V. “Order-Restricted Inference in Multi-Level Step-Stress Life Testing via Generalized Isotonic Regression.” Oral presentation, 23rd Asian Network for Quality (ANQ) Congress 2025, M. S. Ramaiah University of Applied Sciences, M. S. R Nagar, Bengaluru 560054, India.

Sarat Sindhu Mukhopadhyay, and E. V. Gijo “Inference for Proportional Hazard Step-Stress Data Using Inverse Estimation and Fiducial Methods.” Oral presentation, *Eleventh International Conference on Statistics for the Twenty-First Century (ICSTC 2025)*, organized by the International Statistics Fraternity (ISF) in association with the Department of Statistics, University of Kerala, the School of Physical and Mathematical Sciences, University of Kerala, and the American Statistical Association (ASA), India.

Teaching Experience

2021–2026	Reliability, Maintainability and Safety I , Teaching Assistant	<i>ISI, Bangalore</i>
2021–2026	Reliability, Maintainability and Safety II , Teaching Assistant	<i>ISI, Bangalore</i>
2021–2022	Part-time SQC , Teaching Assistant	<i>ISI, Bangalore</i>
2026–2027	Probability , Teaching Assistant	<i>ISI, Bangalore</i>

Mentoring

- 2023-2024 **Pramod Kumar Singh**, Master’s Student, Indian Statistical Institute, Bangalore
- 2024-2025 **Amritha M N**, Master’s Student, Mahatma Gandhi University, India